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1. The set of complex numbers $\mathbb{C} = \{z \mid z = a + bi, a, b \in \mathbb{R}\}$ can be represented geometrically on a plane coordinatized by a horizontal real axis and a vertical imaginary axis, so that each number $a + bi$ corresponds to the point (a, b) . The distance from the point $z = a + bi$ to the origin is then $|z| = \sqrt{a^2 + b^2}$, called the *modulus* of z .
- (a) Show that the set $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ of all points on the unit circle in the complex plane forms a group under complex multiplication (note when multiplying complex numbers that $i^2 = -1$).

We need to show that, under the operation of complex multiplication, S^1 is closed, associative, has an identity element, and has inverses.

Closure: Let $z_1, z_2 \in S^1$. We know that $|z_1| = 1$ and $|z_2| = 1$. Consider the product $z_1 z_2$. We need to show that $|z_1 z_2| = 1$. So,

$$\begin{aligned} |z_1 z_2| &= |z_1| \cdot |z_2| \\ &= 1 \cdot 1 \\ &= 1 \end{aligned}$$

Therefore, $|z_1 z_2| = 1$, so $z_1 z_2 \in S^1$, showing that S^1 is closed under complex multiplication.

Associativity: Let $z_1, z_2, z_3 \in S^1$. We need to show that $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$. Since each z_i is a complex number, we can write $z_1 = a_1 + b_1 i$, $z_2 = a_2 + b_2 i$, and $z_3 = a_3 + b_3 i$ where $a_i, b_i \in \mathbb{R}$. So,

$$\begin{aligned} (z_1 \cdot z_2) \cdot z_3 &= ((a_1 + b_1 i) \cdot (a_2 + b_2 i)) \cdot (a_3 + b_3 i) \\ &= (a_1 a_2 + (a_1 b_2) i + (b_1 a_2) i - b_1 b_2) \cdot (a_3 + b_3 i) \\ &= (a_1 a_2 a_3 + (a_1 b_2 a_3) i + (b_1 a_2 a_3) i - b_1 b_2 a_3) \\ &\quad + ((a_1 a_2 b_3) i - a_1 b_2 b_3 - b_1 a_2 b_3 - (b_1 b_2 b_3) i) \\ &= a_1 [a_2 a_3 + (b_2 a_3) i + (a_2 b_3) i - b_2 b_3] \\ &\quad + b_1 i [(a_2 a_3) + (b_2 a_3) i + (a_2 b_3) i - b_2 b_3] \\ &= a_1 + b_1 i [a_2 a_3 + (b_2 a_3) i + (a_2 b_3) i - b_2 b_3] \\ &= a_1 + b_1 i [a_2 (a_3 + b_3 i) + b_2 i (a_3 + b_3 i)] \\ &= a_1 + b_1 i [a_2 + b_2 i (a_3 + b_3 i)] \\ &= z_1 (z_2 \cdot z_3). \end{aligned}$$

Identity: Consider the element $1 + 0i \in \mathbb{C}$. It must be in S^1 since $|1 + 0i| = \sqrt{1^2 + 0^2} = 1$. Let $z_1 \in S^1$.

$$\begin{aligned} z_1 \cdot 1 + 0i &= z_1 \cdot 1 \\ &= z_1. \end{aligned}$$

Thus, 1 is the identity for S^1 .

Inverses: Let $z_1 = a_1 + b_1i \in S^1$ where $a_1, b_1 \in \mathbb{R}$. Consider its reciprocal $\frac{1}{z_1}$. It must be in S^1 since $|\frac{1}{z_1}| = \frac{|1|}{|z_1|} = \frac{1}{1} = 1$. So,

$$\begin{aligned} z_1 \cdot \frac{1}{z_1} &= (a_1 + b_1i) \cdot \frac{1}{a_1 + b_1i} \\ &= 1. \end{aligned}$$

So, $\frac{1}{z}$ is the inverse for any $z \in S^1$. Therefore S^1 is a group under complex multiplication.

- (b) Let $\mathbb{R}$ be the group of real numbers under addition. Show that the map $\varphi : \mathbb{R} \rightarrow S^1$ given by $\varphi(x) = \cos(x) + i \sin(x)$ is a homomorphism.

We need to show that φ is operation preserving. Let $a, b \in \mathbb{R}$. Then,

$$\begin{aligned} \varphi(a+b) &= \cos(a+b) + i \sin(a+b) \\ &= [\cos(a)\cos(b) - \sin(a)\sin(b)] + i[\sin(a)\cos(b) + \cos(a)\sin(b)] \\ &= [\cos(a)\cos(b) - \sin(a)\sin(b)] + i \sin(a)\cos(b) + i \cos(a)\sin(b) \\ &= \cos(a)[\cos(b) + i \sin(b)] + i \sin(a)[\cos(b) + i \sin(b)] \\ &= (\cos(a) + i \sin(a))(\cos(b) + i \sin(b)) \\ &= \varphi(a)\varphi(b). \end{aligned}$$

- (c) Use the First Isomorphism Theorem on φ to conclude an isomorphism involving the groups $\mathbb{R}$ and S^1 .

By the First Isomorphism Theorem, we know that $\mathbb{R}/\ker\varphi \cong \varphi(\mathbb{R})$. Since the identity of S^1 is 1, the kernel of φ should be all angles θ in the unit circle such that $\cos(\theta) = 1$. Since that is all multiples of 2π , $\ker\varphi = \langle 2\pi \rangle$. Since S^1 is a unit circle, all of S^1 can be reached by some number in $\mathbb{R}$ (i.e. $\varphi(0) = 1$, $\varphi(\frac{\pi}{2}) = i$, $\varphi(\pi) = -1$, $\varphi(\frac{3\pi}{2}) = -i$).

So, $\varphi(\mathbb{R}) = S^1$. Hence, $\boxed{\mathbb{R}/\langle 2\pi \rangle \cong S^1}$

2. (a) Determine all homomorphisms from $\mathbb{Z}_4$ to $\mathbb{Z}_2 \oplus \mathbb{Z}_2$.

We know $\mathbb{Z}_4 = \langle 1 \rangle$. By property 2 of Theorem 10.1, if $\varphi(1) = (a, b)$ then $\varphi(x) = \underbrace{(a, b) + (a, b) + \cdots + (a, b)}_{x \text{ times}} = (xa, xb) \forall x \in \mathbb{Z}_4$. By property 3 of Theorem 10.1 and

Lagrange's Theorem, $|(a, b)|$ must divide $|\mathbb{Z}_4|$ and $|\mathbb{Z}_2 \oplus \mathbb{Z}_2|$. So, $|(a, b)| = 1$ or 2 .

$$\begin{aligned} |(0, 0)| &= \text{lcm}(|0_{\mathbb{Z}_2}|, |0_{\mathbb{Z}_2}|) = 1 \\ |(1, 0)| &= \text{lcm}(|1_{\mathbb{Z}_2}|, |0_{\mathbb{Z}_2}|) = 2 \\ |(0, 1)| &= \text{lcm}(|0_{\mathbb{Z}_2}|, |1_{\mathbb{Z}_2}|) = 2 \\ |(1, 1)| &= \text{lcm}(|1_{\mathbb{Z}_2}|, |1_{\mathbb{Z}_2}|) = 2 \end{aligned}$$

So, there are 4 homomorphisms from $\mathbb{Z}_4$ to $\mathbb{Z}_2 \oplus \mathbb{Z}_2$.

- (b) How many homomorphisms are there from $\mathbb{Z}_{20}$ to $\mathbb{Z}_{10}$? How many of these are onto?

We know $\mathbb{Z}_{20} = \langle 1 \rangle$. By property 2 of Theorem 10.1, if $\varphi(1) = a$ then $\varphi(x) = ax \forall x \in \mathbb{Z}_{20}$. By Lagrange's Theorem and property 3 of Theorem 10.1, $|a|$ must divide $|\mathbb{Z}_{20}|$ and $|\mathbb{Z}_{10}|$. So, $|a| = 1, 2, 5$, or 10 .

$$* |a| = \frac{n}{\gcd(a, n)} \text{ where } a \in \mathbb{Z}_n$$

$$\mathbb{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\Rightarrow \begin{array}{cccc} 1 & 2 & 5 & 10 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 0 & 5 & \{2, 4, 6, 8\} & \{1, 3, 7, 9\} \end{array}$$

So, there are 10 homomorphisms from $\mathbb{Z}_{20}$ to $\mathbb{Z}_{10}$. Of the 10, 4 are onto since only 4 elements in $\mathbb{Z}_{10}$ have order 10.

3. Let G be any subgroup of a dihedral group D_n . For each x in G , define a map φ from G to the multiplicative group $\{+1, -1\}$ by

$$\varphi(x) = \begin{cases} +1 & \text{if } x \text{ is a rotation} \\ -1 & \text{if } x \text{ is a reflection} \end{cases}$$

- (a) Show that φ is a homomorphism.

Let $x_1, x_2 \in G$ be rotations and $y_1, y_2 \in G$ be reflections. Then,

$$\begin{aligned} \varphi(x_1 x_2) &= +1 \\ &= \varphi(x_1) \cdot \varphi(x_2) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \varphi(y_1 y_2) &= +1 \\ &= \varphi(y_1) \cdot \varphi(y_2) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \varphi(x_1 y_1) &= -1 \\ &= \varphi(x_1) \cdot \varphi(y_1) \quad \checkmark \end{aligned}$$

So, φ is a homomorphism.

- (b) What is the kernel of φ ? And explain why this proves the following proposition:

For every subgroup of D_n , either every member of the subgroup is a rotation, or exactly half of the members are rotations.

The kernel of φ contains all elements of G that map to $+1$. Given by the definition of φ , this is all rotations in G . Since $\ker\varphi$ is a subgroup of G , it is also a subgroup of D_n . So, this takes care of the first half of the proposition. Suppose G contained at least one reflection α and $\beta \in G$. If β is a rotation, $\beta\alpha$ is a reflection. If β is a reflection, $\beta\alpha$ is a rotation. This means any rotation in G can reach a reflection and vice versa. This one-to-one correspondence must mean there's an equal number of rotations and reflections in G . Hence, exactly half of the members are rotations.

4. (a) Determine all possible homomorphic images of the symmetric group S_3 .

By Theorem 10.4, this is equivalent to finding the images of S_3/N where N is a normal subgroup of S_3 . By HW 8 # 1, we know that the normal subgroups of S_3 are S_3 itself, the trivial subgroup, and A_3 . Consider the order of the factor groups S_3/N ,

$$|S_3/S_3| = 1 \ (|e|)$$

$$|S_3/\{e\}| = 6 \ (|S_3|)$$

$$|S_3/A_3| = 2 \ (|\mathbb{Z}_2|)$$

So, the homomorphic images of S_3 are S_3 , $\{e\}$, and $\mathbb{Z}_2$.

- (b) Use the First Isomorphism Theorem to prove Theorem 9.4, which says that $G/Z(G) \cong \text{Inn}(G)$ for any group G .

Proof. Let G be a group and $Z(G)$ be the center of G . Let $\varphi : G \rightarrow \text{Inn}(G)$ be a map defined by $\varphi(x) = gxg^{-1} \forall x \in G$ and for some fixed $g \in G$. We first need to show that φ is a homomorphism. If $a, b \in G$, then

$$\begin{aligned} \varphi(a)\varphi(b) &= (gag^{-1})(gbg^{-1}) \\ &= ga(g^{-1}g)bg^{-1} \\ &= gabg^{-1} \\ &= \varphi(ab). \end{aligned}$$

So, φ is a homomorphism. To show that $\varphi(G) = \text{Inn}(G)$, we need to show that φ is onto. Suppose $\varphi_g \in \text{Inn}(G)$. By definition, $\varphi_g(x) = gxg^{-1} \forall x \in G$. So, φ is onto. By Theorem 10.4, $Z(G)$ is the kernel of G since $Z(G)$ is a normal subgroup of G . So, by the First Isomorphism Theorem, $G/Z(G) \cong \text{Inn}(G)$.

5. Suppose $\varphi : G \rightarrow H$ and $\sigma : H \rightarrow K$ are homomorphisms.

- (a) Show that $\sigma\varphi : G \rightarrow K$ is a homomorphism.

Let $g_1, g_2 \in G$. Since φ and σ are homomorphisms,

$$\begin{aligned}
\sigma\varphi(g_1g_2) &= \sigma(\varphi(g_1g_2)) \\
&= \sigma(\varphi(g_1)\varphi(g_2)) \\
&= \sigma(\varphi(g_1))\sigma(\varphi(g_2))
\end{aligned}$$

- (b) How are $\ker \varphi$ and $\ker \sigma\varphi$ related?

Let $g \in \ker \varphi$. This means $\varphi(g) = e_H$. Consider $\sigma\varphi(g)$.

$$\begin{aligned}
\sigma\varphi(g) &= \sigma(\varphi(g)) \\
&= \sigma(e_H) \\
&= e_K.
\end{aligned}$$

So, $g \in \ker \sigma\varphi$. Thus, $\boxed{\ker \varphi \subseteq \ker \sigma\varphi}$.

- (c) If φ and σ are onto and G is finite, describe the index $[\ker \varphi : \ker \sigma\varphi]$ in terms of $|H|$ and $|K|$.

Consider $|G/\ker\varphi|$. By the First Isomorphism Theorem, we know that $|G/\ker\varphi| = |\varphi(G)|$. Since φ is onto, every element of H is reached by an element of G . That is, $|\varphi(G)| = |H|$. So, $|G/\ker\varphi| = |H|$. Similarly, $|H/\ker\sigma| = |K|$ and $|G/\ker\sigma\varphi| = |K|$. So,

$$\begin{aligned}
|\ker \sigma\varphi : \ker \varphi| &= \frac{|\ker \sigma\varphi|}{|\ker \varphi|} \\
&= \frac{\frac{|G|}{|K|}}{\frac{|G|}{|H|}} \\
&= \frac{|G|}{|K|} \cdot \frac{|H|}{|G|} \\
&= \boxed{\frac{|H|}{|K|}}
\end{aligned}$$